

UNSW Mathematics Society Presents
MATH2221/2121 Seminar Part 1



Presented by Steve Jang, James Davidson and Jay Liang

Overview I

1. Part I: Linear ODEs

Linear Differential Operators

Wronskian and Linear Independence

Strategies

Power Series

Bessel & Legendre Equations

2. Dynamical systems

Lipschitz Continuity (2221 only)

Linear dynamical systems

Matrices

Equilibrium points

First Integrals (2221 only)

1. Part I: Linear ODEs

Differential Operators

Definition 1: Linear differential operators

Define the **linear differential operator** L of order m to be

$$\begin{aligned} Lu(x) &= \sum_{j=0}^m a_j(x) \cdot D^j u(x) \\ &= a_m(x) D^m u(x) + a_{m-1}(x) D^{m-1} u(x) + \cdots + a_0(x) u(x), \end{aligned}$$

where $D^j u = \frac{d^j u}{dx^j}$ and $D^0 u = u$.

Differential Operators

Definition 1: Linear differential operators

Define the **linear differential operator** L of order m to be

$$\begin{aligned}Lu(x) &= \sum_{j=0}^m a_j(x) \cdot D^j u(x) \\ &= a_m(x)D^m u(x) + a_{m-1}(x)D^{m-1}u(x) + \cdots + a_0(x)u(x),\end{aligned}$$

where $D^j u = \frac{d^j u}{dx^j}$ and $D^0 u = u$.

Definition 2: Singular ODEs

An ODE is said to be **singular** with respect to $[a, b]$ if the leading coefficient vanishes for any $x \in [a, b]$. E.g. $xu' - u = 0$ is singular on $[0, 1]$.

Homogeneous and inhomogeneous ODEs

Definition 3: Homogeneous ODEs

An ODE is said to be **homogeneous** if the right hand side is 0. That is, we have a differential equation of the form

$$Lu = 0.$$

- **Example:** $u'' + u' - u = 0$.

Homogeneous and inhomogeneous ODEs

Definition 3: Homogeneous ODEs

An ODE is said to be **homogeneous** if the right hand side is 0. That is, we have a differential equation of the form

$$Lu = 0.$$

- **Example:** $u'' + u' - u = 0$.

Definition 4: Inhomogeneous ODEs

An ODE is said to be **inhomogeneous** if the right hand side is not 0. Then we can write the differential equation as

$$Lu = f.$$

- **Example:** $u'' + u' - u = \cos(x)$.

Example 1

Non-exam Question

Solve the following homogeneous linear ODE:

$$u'' + u' - u = 0$$

Example 1

Non-exam Question

Solve the following homogeneous linear ODE:

$$u'' + u' - u = 0$$

Solution.

Suppose that $u = e^{rt}$.

Then,

$$u' = re^{rt}, \text{ and, } u'' = r^2e^{rt}.$$

So, we would have,

$$\begin{aligned} r^2e^{rt} + re^{rt} - e^{rt} &= 0 \\ r^2 + r - 1 &= 0. \end{aligned}$$

Example 1

Solution continued.

We end up with,

$$r_{1,2} = \frac{-1 \pm \sqrt{5}}{2}.$$

Example 1

Solution continued.

We end up with,

$$r_{1,2} = \frac{-1 \pm \sqrt{5}}{2}.$$

Thus, our solutions to the ODE would be,

$$u_1 = e^{r_1 t}, \text{ and, } u_2 = e^{r_2 t}.$$

This gives us our general homogeneous solution of,

$$u_h = Ae^{r_1 t} + Be^{r_2 t}.$$

Example 1

Solution continued.

We end up with,

$$r_{1,2} = \frac{-1 \pm \sqrt{5}}{2}.$$

Thus, our solutions to the ODE would be,

$$u_1 = e^{r_1 t}, \text{ and, } u_2 = e^{r_2 t}.$$

This gives us our general homogeneous solution of,

$$u_h = Ae^{r_1 t} + Be^{r_2 t}.$$

Let's also consider different possible cases, namely, double roots and complex roots.

Initial-value problems (IVP)

Definition 5: Initial-value problems

Consider an m -th order differential equation

$$Lu = f, \quad \text{on } [a, b] \quad (1)$$

along with the values

$$u(a) = v_0, u'(a) = v_1, \dots, u^{(m-1)}(a) = v_{m-1}. \quad (2)$$

The problem (1) with (2) is called an initial-value problem.

Initial-value problems (IVP)

Definition 5: Initial-value problems

Consider an m -th order differential equation

$$Lu = f, \quad \text{on } [a, b] \quad (1)$$

along with the values

$$u(a) = v_0, u'(a) = v_1, \dots, u^{(m-1)}(a) = v_{m-1}. \quad (2)$$

The problem (1) with (2) is called an initial-value problem.

- **Example:** $u' + u = 0, \quad u(0) = 0.$
- **Solution:** $u(x) = e^{-x}.$

Unique solutions

Theorem 1: Unique solution for IVP

If f is continuous on $[a, b]$ and the ODE $Lu = f$ is non-singular on $[a, b]$, then the IVP (1) and (2) has a unique solution.

Unique solutions

Theorem 1: Unique solution for IVP

If f is continuous on $[a, b]$ and the ODE $Lu = f$ is non-singular on $[a, b]$, then the IVP (1) and (2) has a unique solution.

Theorem 2: Homogeneous equation solution

If L is a linear m -th order differential operator and non-singular on $[a, b]$, then the set of all solutions to the homogenous equation $Lu = 0$ on $[a, b]$ forms a vector space of dimension m .

Unique solutions

Theorem 1: Unique solution for IVP

If f is continuous on $[a, b]$ and the ODE $Lu = f$ is non-singular on $[a, b]$, then the IVP (1) and (2) has a unique solution.

Theorem 2: Homogeneous equation solution

If L is a linear m -th order differential operator and non-singular on $[a, b]$, then the set of all solutions to the homogenous equation $Lu = 0$ on $[a, b]$ forms a vector space of dimension m .

- **What does this mean?:** The solution space has a basis of dimension m , with elements $u_1, \dots, u_m$. And so every solution to the homogeneous equation can be written as a linear combination of this basis:

$$u(x) = c_1 u_1(x) + \dots + c_m u_m(x), \quad \forall x \in [a, b].$$

This is called the general solution.

Example 1 Extended

Non-exam Question

Solve the following homogeneous linear IVP:

$$u'' + u' - u = 0, \text{ given that, } u(0) = 0, u'(0) = 2$$

Example 1 Extended

Non-exam Question

Solve the following homogeneous linear IVP:

$$u'' + u' - u = 0, \text{ given that, } u(0) = 0, u'(0) = 2$$

Solution

Recall from before,

$$u_h = Ae^{r_1 t} + Be^{r_2 t}.$$

So making use of the two initial conditions we obtain,

$$u_h(0) = A + B = 0, \text{ and, } u'_h(0) = r_1 A + r_2 B = 2.$$

Example 1 Extended

Non-exam Question

Solve the following homogeneous linear IVP:

$$u'' + u' - u = 0, \text{ given that, } u(0) = 0, u'(0) = 2$$

Solution

Recall from before,

$$u_h = Ae^{r_1 t} + Be^{r_2 t}.$$

So making use of the two initial conditions we obtain,

$$u_h(0) = A + B = 0, \text{ and, } u'_h(0) = r_1 A + r_2 B = 2.$$

So,

$$B = -A, \text{ and thus, } A(r_1 - r_2) = 2.$$

Solving for A and B , we get

$$u_h = \frac{2}{r_1 - r_2} e^{r_1 t} + \frac{2}{r_2 - r_1} e^{r_2 t}.$$

Linear Independence

Definition 6: Linear independence

Let $u_1(x), \dots, u_m(x)$ be functions on some interval $I \subset \mathbb{R}$. We say that $u_1, \dots, u_m$ are **linearly independent** if

$$a_1 u_1(x) + \dots + a_m u_m(x) = 0, \quad \forall x \in I,$$

implies that the constants $a_1, \dots, a_m$ are all 0. Otherwise, we say that that $u_1, \dots, u_m$ are **linearly dependent**.

Definition 7: Wronskian

The **Wronskian** of the functions $u_1, \dots, u_m$ is the $m \times m$ determinant

$$W(x) = W(x; u_1, \dots, u_m) = \det(D^{i-1}u_j).$$

For example, a 3×3 Wronskian is

$$W(x) = W(x; u_1, \dots, u_m) = \det \begin{pmatrix} u_1 & u_2 & u_3 \\ u_1' & u_2' & u_3' \\ u_1'' & u_2'' & u_3'' \end{pmatrix}.$$

Linear Independence of solutions

Theorem 3: Linear independence

Let $u_1, \dots, u_m$ be solutions to the non-singular, linear, homogeneous m -th order ODE $Lu = 0$ on the interval $[a, b]$. Then either

$W(x) = 0$ and the m solutions are linearly dependent,

or

$W(x) \neq 0$ and the m solutions are linearly independent.

Extra Lemmas for Linear Independence and Wronskian

Lemma 1

If $u_1, \dots, u_m$ are linearly dependent over an interval $I \subset \mathbb{R}$ then $W(x; u_1, \dots, u_m) = 0$ for all $x \in I$.

Lemma 2

If $u_1, \dots, u_m$ are solutions to the ODE

$$a_m(x)u^{(m)}(x) + a_{m-1}(x)u^{(m-1)}(x) + \dots + a_0(x)u(x) = 0$$

on an interval $I \subset \mathbb{R}$, then the Wronskian satisfies

$$a_m(x)W'(x) + a_{m-1}(x)W(x) = 0 \quad \forall x \in I.$$

Example 2

Non-exam Question

Show that the homogeneous solutions to the following linear ODE are linearly independent,

$$y'' + 3y = 0.$$

Hint.

First, we must solve the ODE, then next we can make use of the theorem relating the Wronskian to linear independence.

Example 2

Solution.

The characteristic equation we attain would be,

$$r^2 + 3 = 0.$$

Example 2

Solution.

The characteristic equation we attain would be,

$$r^2 + 3 = 0.$$

So, our solutions for r would be,

$$r_{1,2} = \pm i\sqrt{3}.$$

And so our homogeneous solutions would be,

$$u_1 = \cos(\sqrt{3}t) \text{ , and , } u_2 = \sin(\sqrt{3}t).$$

Example 2

Solution continued.

Now computing the Wronskian, W , we get,

$$W(t) = \det \begin{pmatrix} \cos(\sqrt{3}t) & \sin(\sqrt{3}t) \\ -\sqrt{3} \sin(\sqrt{3}t) & \sqrt{3} \cos(\sqrt{3}t) \end{pmatrix}$$

Example 2

Solution continued.

Now computing the Wronskian, W , we get,

$$W(t) = \det \begin{pmatrix} \cos(\sqrt{3}t) & \sin(\sqrt{3}t) \\ -\sqrt{3}\sin(\sqrt{3}t) & \sqrt{3}\cos(\sqrt{3}t) \end{pmatrix}$$

So, $W(t) = \sqrt{3}\cos^2(\sqrt{3}t) + \sqrt{3}\sin^2(\sqrt{3}t) = \sqrt{3} \neq 0$.

Thus, the ODE does indeed have linearly independent homogeneous solutions.

Example 3

MATH2221 2014 Q2. iii)

- b) Prove that if u_1 and u_2 are solutions of a second-order, homogeneous linear differential equation,

$$a_2 u'' + a_1 u' + a_0 u = 0,$$

then their Wronskian, W , satisfies,

$$a_2 W' + a_1 W = 0.$$

- c) Hence determine the form of W when u_1 and u_2 are solutions of the Legendre equation with parameter ν ,

$$(1 - x^2)u'' - 2xu' + \nu(\nu + 1)u = 0.$$

Example 3

Solution.

b)

Firstly note that, as u_1 and u_2 are solutions to the ODE, we have that,

$$a_2 u_1'' + a_1 u_1' + a_0 u_1 = 0 \quad (1)$$

$$a_2 u_2'' + a_1 u_2' + a_0 u_2 = 0. \quad (2)$$

Example 3

Solution.

b)

Firstly note that, as u_1 and u_2 are solutions to the ODE, we have that,

$$a_2 u_1'' + a_1 u_1' + a_0 u_1 = 0 \quad (1)$$

$$a_2 u_2'' + a_1 u_2' + a_0 u_2 = 0. \quad (2)$$

Next, let's compute W and W' .

$$\begin{aligned} W &= \det \begin{pmatrix} u_1 & u_2 \\ u_1' & u_2' \end{pmatrix} \\ &= u_1 u_2' - u_2 u_1' \end{aligned}$$

So,

$$W' = u_2' u_1' + u_1 u_2'' - u_1' u_2' - u_2 u_1''.$$

Example 3

Solution continued.

Now, let's consider the LHS of the equation we have to prove,

$$\begin{aligned}a_2W' + a_1W &= a_2(u_2'u_1' + u_1u_2'' - u_1'u_2' - u_2u_1'') + a_1(u_1u_2' - u_2u_1') \\&= u_1(a_2u_2'' + a_1u_2') - u_2(a_2u_1'' + a_1u_1') \\&= u_1(-a_0u_2) - u_2(-a_0u_1) && \text{(From (1) and (2))} \\&= -a_0u_1u_2 + a_0u_1u_2 \\&= 0.\end{aligned}$$

Example 3

Solution continued.

c)

Using part b) we have that,

$$(1 - x^2)W' - 2xW = 0.$$

This has now turned to simply a first order ODE that can be solved using an integrating factor.

Example 3

Solution continued.

c)

Using part b) we have that,

$$(1 - x^2)W' - 2xW = 0.$$

This has now turned to simply a first order ODE that can be solved using an integrating factor.

Take $\mu(x) = e^{\int \frac{-2x}{1-x^2} dx} = 1 - x^2$.

So, we have that,

$$\frac{d[(1 - x^2)W]}{dx} = 0,$$

Thus,

$$W = \frac{C}{1 - x^2}, \text{ where } C \text{ is a constant.}$$

Reduction of order

Theorem 4: Reduction of order

If we know a solution $u_1(x) \neq 0$ to the ODE

$$u'' + p(x)u' + q(x)u = 0$$

then we can find a second solution

$$u_2(x) = u_1(x) \int \frac{1}{u_1(x)^2 \exp\left(\int p(x)dx\right)} dx.$$

Example 4

MATH2221 2017 Q2. iii)

Show that $u_1 = e^{2x}$ is a solution to the differential equation,

$$(x + 2)u'' - (2x + 5)u' + 2u = 0$$

and find the general solution using the reduction of order method.

Example 4

MATH2221 2017 Q2. iii)

Show that $u_1 = e^{2x}$ is a solution to the differential equation,

$$(x + 2)u'' - (2x + 5)u' + 2u = 0$$

and find the general solution using the reduction of order method.

Solution.

If $u_1 = e^{2x}$, then $u_1' = 2e^{2x}$ and $u_1'' = 4e^{2x}$.

So we have,

$$\begin{aligned}(x + 2)(4e^{2x}) - (2x + 5)(2e^{2x}) + 2e^{2x} &= 4xe^{2x} + 8e^{2x} - 4xe^{2x} - 10e^{2x} \\ &\quad + 2e^{2x} \\ &= 0.\end{aligned}$$

Example 4

Solution continued.

Having verified the first solution, we can now compute the second solution using reduction of order.

So we have,

$$u_2 = u_1 \frac{1}{u_1^2 e^{\int p \, dx}} \, dx,$$

where $p = \frac{-(2x+5)}{x+2}$.

Example 4

Solution continued.

Having verified the first solution, we can now compute the second solution using reduction of order.

So we have,

$$u_2 = u_1 \frac{1}{u_1^2 e^{\int p \, dx}} \, dx,$$

$$\text{where } p = \frac{-(2x+5)}{x+2}.$$

Note that evaluating the inner integral, we have,

$$\begin{aligned} \frac{-(2x+5)}{x+2} \, dx &= -2 - \frac{1}{x+2} \, dx \\ &= -2x - \ln|x+2|. \end{aligned}$$

Example 4

Solution continued.

So, returning to u_2 , we have,

$$\begin{aligned}u_2 &= u_1 \frac{1}{u_1^2 e^{-2x - \ln(x+2)}} dx \\&= e^{2x} \frac{1}{e^{4x} e^{-2x - \ln(x+2)}} dx \\&= e^{2x} (x+2) e^{-2x} dx \\&= e^{2x} e^{-2x} \frac{-(2x+5)}{4} \\&= \frac{-(2x+5)}{4}.\end{aligned}$$

Educated Guessing: Polynomial

Theorem 5

Let $L = p(D)$ be a linear differential operator of order m with constant coefficients. Assume that $p(0) \neq 0$. Then for any integer $r \geq 0$, there exists a unique polynomial u_P of degree r such that $Lu_P = x^r$.

This means that if our linear ODE has a polynomial on the RHS, we should guess a polynomial of the same degree for our particular solution.

Educated Guessing: Exponential

Theorem 6

Let $L = p(D)$ and $\mu \in \mathbb{C}$. If $p(\mu) \neq 0$, then the function

$$u_P(x) = \frac{e^{\mu x}}{p(\mu)}$$

satisfies $Lu_P = e^{\mu x}$.

This means that we should guess a multiple of $e^{\mu x}$ when the RHS of a linear ODE is $e^{\mu x}$ if it is not already a solution of the homogeneous solution.

Example 5

MATH2221 Q1. i) adapted

Find the unique solution $u(x)$ to the ODE,

$$u'' + 4u' + 3u = 9x^2 \text{ where } u(0) = 1 \text{ and } u'(0) = 0.$$

Example 5

MATH2221 Q1. i) adapted

Find the unique solution $u(x)$ to the ODE,

$$u'' + 4u' + 3u = 9x^2 \text{ where } u(0) = 1 \text{ and } u'(0) = 0.$$

Solution.

First solve the homogeneous version of the ODE.

The characteristic equation would be, $r^2 + 4r + 3 = 0$. So the roots would be $r = -1, -3$.

Example 5

MATH2221 Q1. i) adapted

Find the unique solution $u(x)$ to the ODE,

$$u'' + 4u' + 3u = 9x^2 \text{ where } u(0) = 1 \text{ and } u'(0) = 0.$$

Solution.

First solve the homogeneous version of the ODE.

The characteristic equation would be, $r^2 + 4r + 3 = 0$. So the roots would be $r = -1, -3$.

Thus, the general solution to the homogeneous ODE would be,

$$u_h = c_1 e^{-x} + c_2 e^{-3x}, \text{ where } c_1 \text{ and } c_2 \text{ are constants.}$$

Example 5

Solution continued.

The forced term is a polynomial of degree 2, and note that $p(0) = 3 \neq 0$.

Furthermore, there does not exist any polynomial terms in the kernel/homogeneous solution, and so we are free to guess $u_p = Ax^2 + Bx + C$.

Example 5

Solution continued.

The forced term is a polynomial of degree 2, and note that $p(0) = 3 \neq 0$.

Furthermore, there does not exist any polynomial terms in the kernel/homogeneous solution, and so we are free to guess $u_p = Ax^2 + Bx + C$.

So, $u'_p = 2Ax + B$ and $u''_p = 2A$.

Substituting this in, we get,

$$2A + 4(2Ax + B) + 3(Ax^2 + Bx + C) = 9x^2.$$

Example 5

Solution continued.

The forced term is a polynomial of degree 2, and note that $p(0) = 3 \neq 0$.

Furthermore, there does not exist any polynomial terms in the kernel/homogeneous solution, and so we are free to guess $u_p = Ax^2 + Bx + C$.

So, $u'_p = 2Ax + B$ and $u''_p = 2A$.

Substituting this in, we get,

$$2A + 4(2Ax + B) + 3(Ax^2 + Bx + C) = 9x^2.$$

Thus, by comparison of coefficients, we determine that $A = 3$, $B = -8$ and $C = 26/3$.

So the general solution is, $u = c_1e^{-x} + c_2e^{-3x} + 3x^2 - 8x + 26/3$.

Example 5

Solution continued.

Recall that we had arrived at the general homogeneous solution of:

$$u_h = c_1 e^{-x} + c_2 e^{-3x}.$$

And, that we had been given initial conditions of $u(0) = 1$ and $u'(0) = 0$.

Example 5

Solution continued.

Recall that we had arrived at the general homogeneous solution of:

$$u_h = c_1 e^{-x} + c_2 e^{-3x}.$$

And, that we had been given initial conditions of $u(0) = 1$ and $u'(0) = 0$.

Firstly, $u'_h = -c_1 e^{-x} - 3c_2 e^{-3x}$.

So substituting in the initial conditions, we attain the following,

$$c_1 + c_2 = 1 \text{ and, } -c_1 - 3c_2 = 0.$$

Thus, we arrive at $c_1 = 3/2$ and $c_2 = -1/2$.

Example 5

Solution continued.

Recall that we had arrived at the general homogeneous solution of:

$$u_h = c_1 e^{-x} + c_2 e^{-3x}.$$

And, that we had been given initial conditions of $u(0) = 1$ and $u'(0) = 0$.

Firstly, $u'_h = -c_1 e^{-x} - 3c_2 e^{-3x}$.

So substituting in the initial conditions, we attain the following,

$$c_1 + c_2 = 1 \text{ and, } -c_1 - 3c_2 = 0.$$

Thus, we arrive at $c_1 = 3/2$ and $c_2 = -1/2$.

So, the unique solution is ultimately,

$$u = \frac{3}{2}e^{-x} - \frac{1}{2}e^{-3x} + 3x^2 - 8x + 26/3.$$

Example 6

MATH2221 2014 Q2. ii) adapted

Let $p(z) = (z - 1)(z + 2)^2(z^2 + 1)$ and $D = \frac{d}{dx}$.

- a) Write down the general solution u_H of the 5-th order, linear homogeneous ODE $p(D)u = 0$.
- b) Write down the form of a particular solution u_P to the inhomogeneous ODE,

$$p(D)u = e^{-2x} + x^2 + x^2 \cos(x).$$

Example 6

MATH2221 2014 Q2. ii) adapted

Let $p(z) = (z - 1)(z + 2)^2(z^2 + 1)$ and $D = \frac{d}{dx}$.

- a) Write down the general solution u_H of the 5-th order, linear homogeneous ODE $p(D)u = 0$.
- b) Write down the form of a particular solution u_P to the inhomogeneous ODE,

$$p(D)u = e^{-2x} + x^2 + x^2 \cos(x).$$

Solution.

a)

As $p(z)$ is already given, all we need to do is read the roots. We have roots of 1, -2, -2, i , $-i$.

Thus, our general homogeneous solution is,

$$u_h = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x} + c_4 \cos(x) + c_5 \sin(x).$$

Example 6

Solution continued.

b)

Here is the answer, we will dissect it bit by bit.

$$Ax^2e^{-2x} + Bx^2 + Cx + D + x(Ex^2 + Fx + G)\cos(x) + x(Hx^2 + Ix + J)\sin(x).$$

Annihilator method (2221 only)

This method gives us a way to find particular solutions to an inhomogeneous differential equation. Start with an ODE

$$Lu = f(x).$$

We find a differential operator $A(D)$ which **annihilates** $f(x)$, meaning $A(D)f(x) = 0$. For example D^3 annihilates x^2 . We apply this differential operator to both sides of our ODE:

$$A(D)Lu = 0.$$

A solution to this homogeneous differential equation will be a particular solution to the original differential equation.

Example 7

Non-exam Question

Find a particular solution to the ODE

$$y'' - 2y' + y = e^x + \sin(x).$$

Solution.

The LHS differential operator is $p(D) = D^2 - 2D + 1 = (D - 1)^2$. The function e^x is annihilated by $(D - 1)$. The function $\sin(x)$ is annihilated by $(D^2 + 1)$. So the annihilator is $A(D) = (D - 1)(D^2 + 1)$. Apply the annihilator to both sides:

$$A(D)p(D)y(x) = (D - 1)^3(D^2 + 1)y(x) = 0.$$

Solutions to the characteristic equation here is 1 (multiplicity 3), i and $-i$. The particular solution is of the form

$$y_P(x) = c_1 e^x + c_2 x e^x + c_3 x^2 e^x + c_4 \sin(x) + c_5 \cos(x).$$

Example 7

Solution continued.

The first two terms in this particular solution are contained in the homogeneous solution of the original ODE, so we discard them:

$$y_P(x) = c_3 x^2 e^x + c_4 \sin(x) + c_5 \cos(x).$$

Substitute into the original ODE, coefficients are $c_3 = \frac{1}{2}$, $c_4 = 0$ and $c_5 = \frac{1}{2}$. So our particular solution is

$$y_P(x) = \frac{1}{2} x^2 e^x + \frac{1}{2} \cos(x).$$

Variation of Parameters

Theorem 7

If we have a linear, 2nd order, inhomogeneous ODE with leading coefficient 1,

$$Lu = u'' + p(x)u' + q(x)u = f(x).$$

Let u_1, u_2 be a basis for the homogeneous solution space, and let $W(x)$ be the Wronskian $W(x; u_1, u_2)$. Then a particular solution to the inhomogeneous equation is

$$u(x) = v_1(x)u_1(x) + v_2(x)u_2(x),$$

where

$$v_1'(x) = \frac{-u_2(x)f(x)}{W(x)}, \quad \text{and} \quad v_2'(x) = \frac{u_1(x)f(x)}{W(x)}.$$

Example 8

MATH2121 2018 Q1. i)

Use the variation of parameters method to solve

$$y'' - 3y' + 2y = e^x \sin(x).$$

Example 8

MATH2121 2018 Q1. i)

Use the variation of parameters method to solve

$$y'' - 3y' + 2y = e^x \sin(x).$$

Solution.

Homogeneous solution has characteristic equation $t^2 - 3t + 2 = 0$ so $u_1(x) = e^x$ and $u_2(x) = e^{2x}$. The Wronskian is

$$W(x) = \det \begin{pmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{pmatrix} = e^{3x}.$$

Example 8

MATH2121 2018 Q1. i)

Use the variation of parameters method to solve

$$y'' - 3y' + 2y = e^x \sin(x).$$

Solution.

Homogeneous solution has characteristic equation $t^2 - 3t + 2 = 0$ so $u_1(x) = e^x$ and $u_2(x) = e^{2x}$. The Wronskian is

$$W(x) = \det \begin{pmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{pmatrix} = e^{3x}.$$

Then

$$v_1'(x) = \frac{-e^{2x} e^x \sin(x)}{e^{3x}} = -\sin(x),$$

and

$$v_2'(x) = \frac{e^x e^x \sin(x)}{e^{3x}} = e^{-x} \sin(x).$$

Example 8

Solution continued.

Integrating both v_1 and v_2 we have

$$v_1(x) = \cos(x), \quad \text{and} \quad v_2(x) = -\frac{1}{2}e^{-x}(\sin(x) + \cos(x)).$$

Example 8

Solution continued.

Integrating both v_1 and v_2 we have

$$v_1(x) = \cos(x), \quad \text{and} \quad v_2(x) = -\frac{1}{2}e^{-x}(\sin(x) + \cos(x)).$$

Then a particular solution is

$$\begin{aligned} u(x) &= \cos(x)e^x - \frac{1}{2}e^{-x}(\sin(x) + \cos(x))e^{2x} \\ &= \frac{1}{2}e^x(\cos(x) - \sin(x)). \end{aligned}$$

Introduction to Power Series

Theorem 8

Consider a general second-order, linear, homogeneous ODE

$$Lu = a_2(x)u'' + a_1(x)u' + a_0(x)u = 0,$$

or equivalently

$$Lu = u'' + p(x)u' + q(x)u = 0$$

with $p(x) = \frac{a_1(x)}{a_2(x)}$ and $q(x) = \frac{a_0(x)}{a_2(x)}$. Assume that a_0, a_1, a_2 are analytic at 0 (ie. they have local convergent power series), and $a_2(0) \neq 0$. Then p and q are analytic at 0, so we can find power series expansions of both:

$$p(z) = \sum_{k=0}^{\infty} p_k z^k, \quad q(z) = \sum_{k=0}^{\infty} q_k z^k \quad \text{for } |z| < \rho,$$

where $\rho > 0$.

Introduction to Power Series

Theorem 9

If coefficients $p(z)$ and $q(z)$ are analytic for $|z| < \rho$, then the formal power series for the solution $u(z)$ constructed in the previous slide, is also analytic for $|z| < \rho$.

This means that both the functions $p(z)$ and $q(z)$ are analytic, then the power series $u(z)$ is also analytic on the region $|z| < \rho$.

Example 9

MATH2221 2015 Q1. iii)

Consider the ODE

$$(1 + z^2)u'' - zu' - 3u = 0.$$

Find the recurrence relation satisfied by the coefficients A_k in any power series solution:

$$u = \sum_{k=0}^{\infty} A_k z^k.$$

Hence find the solution for which $u(0) = 0$, $u'(0) = 6$.

Note.

$$u'(z) = \sum_{k=1}^{\infty} k A_k z^{k-1}, \quad u''(z) = \sum_{k=2}^{\infty} k(k-1) A_k z^{k-2}.$$

Example 9

Solution.

Substitute into the prior results into the ODE:

$$\begin{aligned} Lu &= \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=2}^{\infty} k(k-1)A_k z^k \\ &\quad - \sum_{k=1}^{\infty} kA_k z^k - \sum_{k=0}^{\infty} A_k z^k \\ &= \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=0}^{\infty} k(k-1)A_k z^k \\ &\quad - \sum_{k=0}^{\infty} kA_k z^k - \sum_{k=0}^{\infty} A_k z^k. \end{aligned}$$

Example 9

Solution.

Substitute into the prior results into the ODE:

$$\begin{aligned} Lu &= \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=2}^{\infty} k(k-1)A_k z^k \\ &\quad - \sum_{k=1}^{\infty} kA_k z^k - \sum_{k=0}^{\infty} A_k z^k \\ &= \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=0}^{\infty} k(k-1)A_k z^k \\ &\quad - \sum_{k=0}^{\infty} kA_k z^k - \sum_{k=0}^{\infty} A_k z^k. \end{aligned}$$

Example 9

Solution continued.

Combine the last three sums.

$$Lu = \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=0}^{\infty} (k^2 - 2k - 3)A_k z^k.$$

Example 9

Solution continued.

Combine the last three sums.

$$Lu = \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=0}^{\infty} (k^2 - 2k - 3)A_k z^k.$$

In the first sum, change k to $k+2$.

$$Lu = \sum_{k=0}^{\infty} (k+1)(k+2)A_{k+2} z^k + \sum_{k=0}^{\infty} (k^2 - 2k - 3)A_k z^k.$$

Example 9

Solution continued.

Combine the last three sums.

$$Lu = \sum_{k=2}^{\infty} k(k-1)A_k z^{k-2} + \sum_{k=0}^{\infty} (k^2 - 2k - 3)A_k z^k.$$

In the first sum, change k to $k+2$.

$$Lu = \sum_{k=0}^{\infty} (k+1)(k+2)A_{k+2} z^k + \sum_{k=0}^{\infty} (k^2 - 2k - 3)A_k z^k.$$

Finally, we can combine the sums.

$$Lu = \sum_{k=0}^{\infty} (k+1) [(k+2)A_{k+2} + (k-3)A_k] z^k.$$

Example 9

Solution continued.

Since $Lu = 0$, we need $(k+2)A_{k+2} + (k-3)A_k = 0$. Rearranging,

$$A_{k+2} = -\frac{k-3}{k+2}A_k.$$

Example 9

Solution continued.

Since $Lu = 0$, we need $(k+2)A_{k+2} + (k-3)A_k = 0$. Rearranging,

$$A_{k+2} = -\frac{k-3}{k+2}A_k.$$

Since $A_5 = -\frac{3-3}{3+2}A_3 = 0$, then all odd terms past A_5 are zero. Also $A_1 = 6$ and $A_3 = 4$.

The even terms start at $A_0 = 0$ so all even terms past this are zero. Hence

$$u(z) = 6z + 4z^3.$$

Cauchy-Euler ODEs

Definition 8: Singular ODEs

Singular ODEs are ODEs where the leading coefficient vanishes for one or more values of $x \in (a, b)$.

Definition 9: Cauchy-Euler ODE

A second-order **Cauchy-Euler ODE** has the form

$$Lu = ax^2u'' + bxu' + cu = f(x),$$

where a, b, c are constants with $a \neq 0$.

Cauchy-Euler ODEs

Definition 8: Singular ODEs

Singular ODEs are ODEs where the leading coefficient vanishes for one or more values of $x \in (a, b)$.

Definition 9: Cauchy-Euler ODE

A second-order **Cauchy-Euler ODE** has the form

$$Lu = ax^2u'' + bxu' + cu = f(x),$$

where a, b, c are constants with $a \neq 0$.

This is singular at $x = 0$. Applying this differential operator L to x^r ,

$$Lx^r = [ar(r-1) + br + c]x^r,$$

we can see that x^r is a solution to the homogeneous equation $Lu = 0$ iff

$$ar(r-1) + br + c = 0. \quad \leftarrow \text{Indicial Equation}$$

Cauchy-Euler ODEs

Lemma 3

Suppose there are distinct solutions r_1, r_2 to the equation $ar(r-1) + br + c = 0$. That is, $r_1 \neq r_2$. Then the general solution of the homogeneous equation $Lu = 0$ is

$$u(x) = C_1x^{r_1} + C_2x^{r_2}, \quad x > 0.$$

Cauchy-Euler ODEs

Lemma 3

Suppose there are distinct solutions r_1, r_2 to the equation $ar(r-1) + br + c = 0$. That is, $r_1 \neq r_2$. Then the general solution of the homogeneous equation $Lu = 0$ is

$$u(x) = C_1x^{r_1} + C_2x^{r_2}, \quad x > 0.$$

Lemma 4

Suppose there is one solution r_1 to $ar(r-1) + br + c = 0$. Then the general solution to the homogeneous equation $Lu = 0$ is

$$C_1x^{r_1} + C_2x^{r_1} \log(x), \quad x > 0.$$

Example 10

MATH2121 2016 Q2. i)

Find the general solution of the Cauchy-Euler ODE

$$2x^2y'' + 7xy' + 3y = 13x^{1/4}, \quad x > 0.$$

Example 10

MATH2121 2016 Q2. i)

Find the general solution of the Cauchy-Euler ODE

$$2x^2y'' + 7xy' + 3y = 13x^{1/4}, \quad x > 0.$$

Note.

This question turns out to be more than just an ordinary Cauchy-Euler problem. It is inhomogeneous with a forced term that does not correspond to any of the 'Educated Guessing' methods that we went through earlier. We shall see that instead this question interestingly makes use of the Variation of Parameters technique.

Example 10

Solution.

First note that we can solve the homogeneous version of this problem by the Indicial Equation,

$$2r(r - 1) + 7r + 3 = 0$$

$$2r^2 + 5r + 3 = 0$$

$$(2r + 3)(r + 1) = 0.$$

Thus, $r = -3/2$ or -1 .

And so, the homogeneous solutions are,

$$y_1 = x^{-3/2} \text{ and } y_2 = x^{-1}.$$

Example 10

Solution continued.

Next, to find a particular solution, we can apply variation of parameters.

Recall that we have,

$$v_1'(x) = \frac{-u_2(x)f(x)}{W(x)}, \quad \text{and} \quad v_2'(x) = \frac{u_1(x)f(x)}{W(x)}.$$

Example 10

Solution continued.

Next, to find a particular solution, we can apply variation of parameters.

Recall that we have,

$$v_1'(x) = \frac{-u_2(x)f(x)}{W(x)}, \quad \text{and} \quad v_2'(x) = \frac{u_1(x)f(x)}{W(x)}.$$

First, computing the Wronskian,

$$\begin{aligned} W(x) &= \det \begin{pmatrix} x^{-3/2} & x^{-1} \\ \frac{-3}{2}x^{-5/2} & -x^{-2} \end{pmatrix} \\ &= -x^{-7/2} + \frac{3}{2}x^{-7/2} \\ &= \frac{1}{2}x^{-7/2}. \end{aligned}$$

Example 10

Solution continued.

Now before we apply variation of parameters, we must first modify our initial problem such that its in a suitable form for variation of parameters.

$$2x^2y'' + 7xy' + 3y = 13x^{1/4} \iff y'' + \frac{7}{2x}y' + \frac{3}{2x^2}y = \frac{13}{2}x^{-7/4}.$$

Example 10

Solution continued.

Now before we apply variation of parameters, we must first modify our initial problem such that its in a suitable form for variation of parameters.

$$2x^2y'' + 7xy' + 3y = 13x^{1/4} \iff y'' + \frac{7}{2x}y' + \frac{3}{2x^2}y = \frac{13}{2}x^{-7/4}.$$

Now, let's compute $v_1(x)$,

$$\begin{aligned} v_1(x) &= \frac{-y_2(x)f(x)}{W(x)} dx \\ &= \frac{-x^{-1}\left(\frac{13}{2}x^{-7/4}\right)}{\frac{1}{2}x^{-7/2}} dx \\ &= 13x^{3/4} dx. \end{aligned}$$

Example 10

Solution continued.

So, $v_1(x) = 13(\frac{4}{7}x^{7/4}) = \frac{52}{7}x^{7/4}$.

Similarly, we can compute $v_2(x)$, and this would ultimately lead us to,

$$y_P(x) = -8x^{1/4}.$$

So,

$$y(x) = C_1x^{-1} + C_2x^{-3/2} - 8x^{1/4}.$$

Frobenius normal form

Definition 10: Frobenius normal form

A frequent form of ODE that appears in many applications can be written in **Frobenius normal form**:

$$z^2 u'' + zP(z)u' + Q(z)u = 0,$$

where $P(z)$ and $Q(z)$ are analytic at 0. Let $P_0 = P(0)$ and $Q_0 = Q(0)$, and define a series F as

$$F(z) = z^r \sum_{k=0}^{\infty} A_k z^k,$$

where r is the solution(s) of $r(r-1) + P_0 r + Q_0 = 0$.

Frobenius normal form

Definition 10: Frobenius normal form

A frequent form of ODE that appears in many applications can be written in **Frobenius normal form**:

$$z^2 u'' + zP(z)u' + Q(z)u = 0,$$

where $P(z)$ and $Q(z)$ are analytic at 0. Let $P_0 = P(0)$ and $Q_0 = Q(0)$, and define a series F as

$$F(z) = z^r \sum_{k=0}^{\infty} A_k z^k,$$

where r is the solution(s) of $r(r-1) + P_0 r + Q_0 = 0$.

Note. If $r_1 \neq r_2$, then $f(z; r_1)$ is a solution to the Frobenius normal form ODE. If $r_1 - r_2$ is **not** a whole number, then a second linearly independent solution is $F(z; r_2)$.

Example 11

MATH2221 2019 Q2. ii)

Consider the following ODE

$$x^2 y'' + xy' + (x^2 - \frac{1}{9})y = 0.$$

- a) Determine the indicial equation and its roots to the ODE
Frobenius method is applied to to $x_0 = 0$.
- b) Write down the first 3 non-zero terms of the power series solution about the larger one of the two roots.

Example 11

Solution.

a)

Note that $P(x) = 1$ and $Q(x) = (x^2 - \frac{1}{9})$.

So, $P(0) = 1$ and $Q(0) = -\frac{1}{9}$.

Example 11

Solution.

a)
Note that $P(x) = 1$ and $Q(x) = (x^2 - \frac{1}{9})$.
So, $P(0) = 1$ and $Q(0) = -\frac{1}{9}$.

So our approximating Cauchy-Euler equation is,

$$x^2 y'' + xy' - \frac{1}{9}y = 0.$$

Thus, our indicial equation is,

$$r(r-1) + r - \frac{1}{9} = 0.$$

Example 11

Solution continued.

Finding roots of our indicial equation,

$$\begin{aligned}r(r-1) + r - \frac{1}{9} &= r^2 - r + r - \frac{1}{9} \\&= r^2 - \frac{1}{9} \\&= \left(r - \frac{1}{3}\right)\left(r + \frac{1}{3}\right)\end{aligned}$$

And so, $r_1 = \frac{1}{3}$ and $r_2 = -\frac{1}{3}$.

Example 11

Solution continued.

b)

Now, as the question specifies that we work with the larger root, we will work with $r = \frac{1}{3}$.

Example 11

Solution continued.

b)

Now, as the question specifies that we work with the larger root, we will work with $r = \frac{1}{3}$.

Let $y = \sum_{k=0}^{\infty} A_k x^{k+r}$.

Then we have that,

$$y' = \sum_{k=0}^{\infty} (k+r) A_k x^{k+r-1} \text{ and, } y'' = \sum_{k=0}^{\infty} (k+r)(k+r-1) A_k x^{k+r-2}.$$

Example 11

Solution continued.

We now substitute into the given equation and manipulate the expression until we can form a recurrence relation.

$$\begin{aligned}x^2y'' + xy' - \frac{1}{9}y + x^2y &= \sum_{k=0}^{\infty} (k+r)(k+r-1)A_kx^{k+r} + \\&\quad \sum_{k=0}^{\infty} (k+r)A_kx^{k+r} - \frac{1}{9} \sum_{k=0}^{\infty} A_kx^{k+r} + \sum_{k=0}^{\infty} A_kx^{k+r+2} \\&= \sum_{k=0}^{\infty} A_kx^{k+r} [(k+r)(k+r-1) + (k+r) - \frac{1}{9}] + \sum_{k=0}^{\infty} A_kx^{k+r+2} \\&= \sum_{k=0}^{\infty} A_kx^{k+r} [(k+r)^2 - \frac{1}{9}] + \sum_{k=2}^{\infty} A_{k-2}x^{k+r}.\end{aligned}$$

Example 11

Solution continued.

We now substitute into the given equation and manipulate the expression until we can form a recurrence relation.

$$\begin{aligned}x^2y'' + xy' - \frac{1}{9}y + x^2y &= \sum_{k=0}^{\infty} (k+r)(k+r-1)A_kx^{k+r} + \\&\sum_{k=0}^{\infty} (k+r)A_kx^{k+r} - \frac{1}{9}\sum_{k=0}^{\infty} A_kx^{k+r} + \sum_{k=0}^{\infty} A_kx^{k+r+2} \\&= \sum_{k=0}^{\infty} A_kx^{k+r}[(k+r)^2 - \frac{1}{9}] + \sum_{k=2}^{\infty} A_{k-2}x^{k+r}.\end{aligned}$$

Example 11

Solution continued.

We now substitute into the given equation and manipulate the expression until we can form a recurrence relation.

$$\begin{aligned}x^2y'' + xy' - \frac{1}{9}y + x^2y &= \sum_{k=0}^{\infty} (k+r)(k+r-1)A_kx^{k+r} + \\&\quad \sum_{k=0}^{\infty} (k+r)A_kx^{k+r} - \frac{1}{9} \sum_{k=0}^{\infty} A_kx^{k+r} + \sum_{k=0}^{\infty} A_kx^{k+r+2} \\&= \sum_{k=0}^{\infty} A_kx^{k+r} [(k+r)(k+r-1) + (k+r) - \frac{1}{9}] + \sum_{k=0}^{\infty} A_kx^{k+r+2} \\&= \sum_{k=0}^{\infty} A_kx^{k+r} [(k+r)^2 - \frac{1}{9}] + \sum_{k=2}^{\infty} A_{k-2}x^{k+r}.\end{aligned}$$

Example 11

Solution continued.

Now note that for the first summation, the $k = 0, 1$ terms becomes,

$$A_0 x^r \left[r^2 - \frac{1}{9} \right] + A_1 x^{r+1} \left[(r+1)^2 - \frac{1}{9} \right].$$

Given that we have $r = \frac{1}{3}$, we get the following for A_0 and A_1 ,

A_0 is a free parameter, and $A_1 = 0$.

Example 11

Solution continued.

Now note that for the first summation, the $k = 0, 1$ terms becomes,

$$A_0 x^r \left[r^2 - \frac{1}{9} \right] + A_1 x^{r+1} \left[(r+1)^2 - \frac{1}{9} \right].$$

Given that we have $r = \frac{1}{3}$, we get the following for A_0 and A_1 ,

A_0 is a free parameter, and $A_1 = 0$.

So, having removed the $k = 0, 1$ terms, we are left with the following sum,

$$\sum_{k=2}^{\infty} A_k x^{k+r} \left[(k+r)^2 - \frac{1}{9} \right] + A_{k-2} x^{k+r}.$$

Example 11

Solution continued.

Thus, we ultimately get the recurrence relation,

$$A_k = -\frac{A_{k-2}}{(k+r)^2 - \frac{1}{9}}.$$

Example 11

Solution continued.

Thus, we ultimately get the recurrence relation,

$$A_k = -\frac{A_{k-2}}{(k+r)^2 - \frac{1}{9}}.$$

Noting that as $A_1 = 0$, we will only have even terms in our series.

Therefore, the first 3 non-zero terms will indeed be

$$A_0x^{1/3}, A_2x^{2+1/3}, A_4x^{4+1/3}.$$

Thus, the answer is ultimately,

$$y = x^{\frac{1}{3}}\left(1 - \frac{3}{16}x^2 + \frac{9}{896}x^4 - \dots\right).$$

Bessel function

The **Bessel equation with parameter ν** is:

$$z^2 u'' + zu' + (z^2 - \nu^2)u = 0.$$

This ODE is in Frobenius normal form, with indicial polynomial:

$$I(r) = (r + \nu)(r - \nu),$$

and we seek a series solution:

$$u(z) = \sum_{k=0}^{\infty} A_k z^{k+r}.$$

We assume $\operatorname{Re}(\nu) \geq 0$, so $r_1 = \nu$ and $r_2 = -\nu$.

Bessel function

With the normalisation:

$$A_0 = \frac{1}{2^\nu \Gamma(1 + \nu)}$$

the series solution is called the **Bessel function of order ν** and is denoted:

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(1 + \nu)} \left[1 - \frac{(z/2)^\nu}{1 + \nu} + \frac{(z/2)^4}{2!(1 + \nu)(2 + \nu)} - \dots \right].$$

Bessel function

With the normalisation:

$$A_0 = \frac{1}{2^\nu \Gamma(1 + \nu)}$$

the series solution is called the **Bessel function of order ν** and is denoted:

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(1 + \nu)} \left[1 - \frac{(z/2)^\nu}{1 + \nu} + \frac{(z/2)^4}{2!(1 + \nu)(2 + \nu)} - \dots \right].$$

And from the functional equation $\Gamma(1 + z) = z\Gamma(z)$:

$$J_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k+\nu}}{k! \Gamma(k + 1 + \nu)}$$

Bessel function

If ν is not an integer, then a second linearly independent, solution is:

$$J_{-\nu}(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k-\nu}}{k! \Gamma(k+1-\nu)}.$$

Bessel function

If ν is not an integer, then a second linearly independent, solution is:

$$J_{-\nu}(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k-\nu}}{k! \Gamma(k+1-\nu)}.$$

For an integer $\nu = n \in \mathbb{Z}$, since $\Gamma(n+1) = n!$, we have:

$$J_n(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k+n}}{k! (k+n)!}.$$

Also, since $\frac{1}{\Gamma(z)} = 0$ for $z = 0, -1, -2, \dots$, we find that J_n and J_{-n} are linearly dependent; in fact:

$$J_{-n}(z) = (-1)^n J_n(z).$$

Legendre equation (2221 only)

The **Legendre equation** with parameter ν is:

$$(1 - z^2)u'' - 2zu' + \nu(\nu + 1)u = 0.$$

This ODE is not singular at $z = 0$, so the solution has an ordinary Taylor series expansion:

$$u = \sum_{k=0}^{\infty} A_k z^k.$$

Legendre equation (2221 only)

The **Legendre equation** with parameter ν is:

$$(1 - z^2)u'' - 2zu' + \nu(\nu + 1)u = 0.$$

This ODE is not singular at $z = 0$, so the solution has an ordinary Taylor series expansion:

$$u = \sum_{k=0}^{\infty} A_k z^k.$$

The A_k must satisfy:

$$(k + 1)(k + 2)A_{k+2} - [k(k + 1) - \nu(\nu + 1)]A_k = 0.$$

The recurrence relation is:

$$A_{k+2} = \frac{(k - \nu)(k + \nu + 1)}{(k + 1)(k + 2)} A_k \quad \text{for } k \geq 0.$$

2. Dynamical systems

Dynamical systems

State variables are natural variables $x_k = x_k(t)$ which depend on a single independent variable (e.g. time). Many natural processes may be modelled as **dynamical systems** described by state variables. The state of a system at a given time is described by a state vector

$$\mathbf{x} = \mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

composed of one or more state variables at that instant.

Autonomous and non-autonomous ODEs

For simplicity, we assume each state variable is dependent only on t .

Definition

A system of ODEs of the form

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} dx_1/dt \\ dx_2/dt \\ \vdots \\ dx_n/dt \end{bmatrix} = \mathbf{F}(\mathbf{x})$$

is said to be **autonomous**. If $\mathbf{F}$ explicitly depends upon t , as in

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t)$$

then the ODE is said to be **non-autonomous**.

Autonomous and non-autonomous ODEs

Each solution $\mathbf{x}$ of the non-autonomous ODE

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t)$$

corresponds to a solution $\mathbf{y}$ of the autonomous ODE

$$\frac{d\mathbf{y}}{dt} = \begin{bmatrix} \mathbf{F}(\mathbf{x}, t) \\ 1 \end{bmatrix} = \mathbf{G}(\mathbf{y}).$$

We therefore focus on the theory of autonomous dynamical systems.

It is worth noting also that $\mathbf{F}$ and $\mathbf{G}$ as described are vector fields in the phase space (i.e. the set of all permissible states)!

Lipschitz continuity (2221 only)

To be able to say when a solution to a dynamical system exists and is unique requires the notion of **Lipschitz continuity**.

Definition

A constant $L \in \mathbb{R}$ is a **Lipschitz constant** for $f : [a, b] \rightarrow \mathbb{R}$ if

$$|f(x) - f(y)| \leq L |x - y| \quad \forall x, y \in [a, b].$$

If such an L exists for f , then f is **Lipschitz continuous**.

Lipschitz continuity (2221 only)

Theorem

If a function is Lipschitz continuous, it is uniformly continuous.

Theorem

If a function is continuously differentiable on a *compact* interval, then it is Lipschitz continuous on that interval.

The contrapositive of the first statement may be useful for disproving Lipschitz continuity, and the second statement may be useful for showing something *is* Lipschitz continuous.

Lipschitz continuity (2221 only)

Example 11

Is $f(x) = \tan x$ Lipschitz on $I_1 = [-\pi/3, \pi/3]$? $I_2 = (-\pi/2, \pi/2)$?

Since f is

Lipschitz continuity (2221 only)

Example 11

Is $f(x) = \tan x$ Lipschitz on $I_1 = [-\pi/3, \pi/3]$? $I_2 = (-\pi/2, \pi/2)$?

Since f is continuously differentiable on the compact interval I_1 , it is immediately Lipschitz on I_1 .

Lipschitz continuity (2221 only)

Example 11

Is $f(x) = \tan x$ Lipschitz on $I_1 = [-\pi/3, \pi/3]$? $I_2 = (-\pi/2, \pi/2)$?

Since f is continuously differentiable on the compact interval I_1 , it is immediately Lipschitz on I_1 . However it fails to be Lipschitz on I_2 , as

Lipschitz continuity (2221 only)

Example 11

Is $f(x) = \tan x$ Lipschitz on $I_1 = [-\pi/3, \pi/3]$? $I_2 = (-\pi/2, \pi/2)$?

Since f is continuously differentiable on the compact interval I_1 , it is immediately Lipschitz on I_1 . However it fails to be Lipschitz on I_2 , as no Lipschitz constant exists: $\tan$ is unbounded near $\pi/2$.

(What went wrong? I_2 is not compact like I_1 was!)

Lipschitz Continuity (2221 only)

Example 11

In the previous example, find a Lipschitz constant for f on I_1 .

For any $x, y \in I_1$ with $x < y$, the MVT ensures the existence of some $c \in (x, y)$ such that

$$|\tan x - \tan y| = |\sec^2 c| |x - y|.$$

Lipschitz Continuity (2221 only)

Example 11

In the previous example, find a Lipschitz constant for f on I_1 .

For any $x, y \in I_1$ with $x < y$, the MVT ensures the existence of some $c \in (x, y)$ such that

$$|\tan x - \tan y| = |\sec^2 c| |x - y|.$$

There are a few ways we could find an upper bound for $|\sec^2(c)|$: the most direct one is to notice that since $c \in I_2$, we have

$$\cos c \leq \cos \frac{\pi}{3} = \frac{1}{2} \implies \sec c \leq 2 \implies \sec^2 c \leq 4.$$

So a suitable Lipschitz constant is $L = 4$.

Lipschitz continuous vector fields (2221 only)

We extend the definition of Lipschitz continuity to vector fields.

Definition 11

A vector field $\mathbf{F} : S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **Lipschitz continuous** on S if there exists some constant $L \in \mathbb{R}$ such that

$$\|\mathbf{F}(\mathbf{x}) - \mathbf{F}(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in S$$

where $\|\cdot\|$ is the usual Euclidean norm on $\mathbb{R}^n$. A time-dependent vector field $\mathbf{F} = \mathbf{F}(\mathbf{x}, t)$ is **Lipschitz in $\mathbf{x}$** if

$$\|\mathbf{F}(\mathbf{x}, t) - \mathbf{F}(\mathbf{y}, t)\| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y}, t.$$

Existence and uniqueness theorem (2221 only)

Theorem 10

Let $r > 0, \tau > 0$ and suppose

$$S = \{(\mathbf{x}, t) : \|\mathbf{x} - \mathbf{x}_0\| \leq r, |t - t_0| \leq \tau\}.$$

If $\mathbf{F}(\mathbf{x}, t)$ is Lipschitz in $\mathbf{x}$ and $\|\mathbf{F}(\mathbf{x}, t)\| \leq M$ for $(\mathbf{x}, t) \in S$, then

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t), \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

has a unique (C^1) solution $\mathbf{x}(t)$ for $|t| \leq \min(r/M, \tau)$.

This result is quite technical, and probably isn't too useful to know (unless you're trying to get 100 in the course or something).

NB: C^1 = once continuously differentiable.

Linear systems of ODEs

Definition 12

We say that the $n \times n$, first-order system of ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t)$$

is **linear** if the RHS has the form

$$\mathbf{F}(\mathbf{x}, t) = A(t)\mathbf{x} + \mathbf{b}(t)$$

for some $n \times n$ matrix-valued function $A(t) = [a_{i,j}(t)]$ and a vector-valued function $\mathbf{b}(t) = [b_i(t)]$.

This system is autonomous when A and $\mathbf{b}$ are constant.

Global existence and uniqueness

We have a stronger existence result in the linear case.

Theorem 11

If A and $\mathbf{b}$ are continuous^a on $[0, T]$, then the linear IVP

$$\frac{d\mathbf{x}}{dt} = A(t)\mathbf{x} + \mathbf{b}(t), \quad \mathbf{x}(0) = \mathbf{x}_0$$

has a unique solution $\mathbf{x}(t)$ for $0 \leq t \leq T$.

^aFormally, continuous as vector and matrix-valued functions!

A special case

It is much easier to work with the special case when $A(t) = A$ is a constant $n \times n$ matrix and $\mathbf{b}(t) = \mathbf{0}$: the resulting system is

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}.$$

The general solution (**assuming A is full rank!**) to this system is

$$\mathbf{x}(t) = \sum_{i=1}^n c_i e^{\lambda_i t} \mathbf{v}_i,$$

where λ_i is the i th eigenvalue with corresponding eigenvector $\mathbf{v}_i$ and $c_1, \dots, c_n$ are constants. So, solving such systems amounts to finding the eigenpairs of A .

Initial-valued system

Recall that for an $n \times n$ complex matrix A , its exponential is

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots.$$

Consider the initial-valued system

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}, \quad \mathbf{x}(0) = \mathbf{x}_0.$$

Then we can write the solution as a matrix exponential

$$\mathbf{x}(t) = e^{tA}\mathbf{x}_0.$$

(Exercise for MATH2221: prove this again!)

Example

Example (MATH2121 2016 T2 2.iii)

Show that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^A\mathbf{v} = e^\lambda\mathbf{v}$.

If $A\mathbf{v} = \lambda\mathbf{v}$ then

$$e^A\mathbf{v}$$

Example

Example (MATH2121 2016 T2 2.iii)

Show that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^A\mathbf{v} = e^\lambda\mathbf{v}$.

If $A\mathbf{v} = \lambda\mathbf{v}$ then

$$e^A\mathbf{v} = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \mathbf{v}$$

Example

Example (MATH2121 2016 T2 2.iii)

Show that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^A\mathbf{v} = e^\lambda\mathbf{v}$.

If $A\mathbf{v} = \lambda\mathbf{v}$ then

$$e^A\mathbf{v} = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \mathbf{v} = \sum_{k=0}^{\infty} \frac{1}{k!} \lambda^k \mathbf{v}$$

Example

Example (MATH2121 2016 T2 2.iii)

Show that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^A\mathbf{v} = e^\lambda\mathbf{v}$.

If $A\mathbf{v} = \lambda\mathbf{v}$ then

$$e^A\mathbf{v} = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \mathbf{v} = \sum_{k=0}^{\infty} \frac{1}{k!} \lambda^k \mathbf{v} = \underbrace{\left(\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \right)}_{=e^\lambda} \mathbf{v} = e^\lambda \mathbf{v}.$$

Calculating the matrix exponential

Calculating e^{tA} involves calculating A^k for $k = 1, 2, \dots$, which is in general a pain. However, the matrix exponential is not so hard to calculate in the following situations:

- If A is nilpotent (i.e. $A^n = \mathbf{0}$ for some integer $n > 0$), then

$$e^A = \sum_{k=0}^{n-1} \frac{A^k}{k!} + \underbrace{\sum_{k=n}^{\infty} \frac{A^k}{k!}}_{=0} = \sum_{k=0}^{n-1} \frac{A^k}{k!}.$$

- If A is diagonalisable, then we can avoid calculating A^k completely!
- If A can be written as a sum of two commuting matrices B and C (i.e. $BC = CB$), then we can write $e^A = e^B e^C$. We can use the above two methods to find e^B and e^C , which may be easier if e^B and e^C are easily computable.

Diagonalising a matrix

Definition 13

An $n \times n$ complex matrix A is **diagonalisable** if there exists a non-singular $n \times n$ matrix Q such that $Q^{-1}AQ$ is diagonal.

Theorem 12

An $n \times n$ complex matrix A is diagonalisable if and only if there exists an eigenbasis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ for $\mathbb{C}^n$. Moreover $\Lambda = Q^{-1}AQ$ with

$$Q = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n) \quad \text{and} \quad \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix},$$

where λ_i is the eigenvalue corresponding to eigenvector $\mathbf{v}_i$.

Matrix powers

Diagonalisation is useful for calculating matrix powers, which you can then use to calculate matrix exponentials. For any integer k , we have

$$A^k = \overbrace{(Q\Lambda Q^{-1})(Q\Lambda Q^{-1}) \cdots (Q\Lambda Q^{-1})}^{k \text{ times}} = Q\Lambda^k Q^{-1},$$

and Λ^k admits the easily computable form

$$\Lambda^k = \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix}.$$

Exponential of a diagonalisable matrix

Theorem 13

If A has diagonalisation $Q\Lambda Q^{-1}$, then

$$e^A = Qe^\Lambda Q^{-1} \quad \text{and} \quad e^\Lambda = \begin{pmatrix} e^{\lambda_1} & & \\ & \ddots & \\ & & e^{\lambda_n} \end{pmatrix}.$$

Since we often need to find e^{tA} rather than e^A , this implies that

$$e^{tA} = Qe^{t\Lambda} Q^{-1} \quad \text{and} \quad e^{t\Lambda} = \begin{pmatrix} e^{t\lambda_1} & & \\ & \ddots & \\ & & e^{t\lambda_n} \end{pmatrix}.$$

Upshot: if you can diagonalise, you can exponentiate matrices!

Equilibrium points

Definition 14

We say that $\mathbf{a} \in \mathbb{R}^n$ is an **equilibrium point** for a dynamical system determined by vector field $\mathbf{F}$ if $\mathbf{F}(\mathbf{a}) = \mathbf{0}$.

Suppose $\mathbf{a}$ is an equilibrium point for the system $\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x})$. Consider the initial-valued system

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{a}.$$

Then the solution is the constant function $\mathbf{x}(t) = \mathbf{a}$.

Stable equilibrium

Definition 15

An equilibrium point $\mathbf{a}$ is **stable** if for every $\epsilon > 0$, there exists $\delta > 0$ such that whenever $\|\mathbf{x}_0 - \mathbf{a}\| < \delta$, the solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0$$

satisfies

$$\|\mathbf{x}(t) - \mathbf{a}\| < \epsilon \quad \forall t > 0.$$

Intuitively: if a solution starts close enough to the stable equilibrium point, then they will remain close to the stable equilibrium point.

Asymptotic stability

This is a stronger form of stability, on a particular subset of $\mathbb{R}^n$.

Definition 16

Let D be an **open** subset of $\mathbb{R}^n$ that contains an equilibrium point $\mathbf{a}$. We say that $\mathbf{a}$ is **asymptotically stable** in D if $\mathbf{a}$ is stable, and whenever $\mathbf{x}_0 \in D$ the solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \mathbf{x}(0) = \mathbf{x}_0$$

satisfies

$$\mathbf{x}(t) \rightarrow \mathbf{a} \quad \text{as } t \rightarrow \infty.$$

We call D a domain of attraction for $\mathbf{a}$.

Intuitively: not only do the solutions stay close to the stable equilibrium point, but they also approach the equilibrium point as t goes to infinity.

Linear constant case

Consider the linear system

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x} + \mathbf{b}, \quad \mathbf{x}(0) = \mathbf{x}_0,$$

where A is non-singular. Then the unique equilibrium point of this system is $\mathbf{a} = -A^{-1}\mathbf{b}$, and the solution to the system is

$$\mathbf{x}(t) = \mathbf{a} + e^{tA} (\mathbf{x}_0 - \mathbf{a}).$$

Linear constant case

Theorem 14

Consider the previous linear constant coefficient system. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A . The equilibrium point $\mathbf{a} = -A^{-1}\mathbf{b}$ is

1. **stable** if and only if $\operatorname{Re}(\lambda_j) \leq 0$ for all j .
2. **asymptotically stable** if and only if $\operatorname{Re}(\lambda_j) < 0$ for all j (with domain of attraction $\mathbb{R}^n$).

If it is neither, then we call it **unstable**.

Classifying 2D linear systems

Given the system

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

with A non-singular, the general solution will be one of two things:

- If the eigenvalues λ_1, λ_2 are real with linearly independent eigenvectors, then

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2.$$

If $\lambda_1 = \lambda_2$, then $\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1$ instead.

- If the eigenvalues are complex conjugates, then

$$\mathbf{x}(t) = c_1 \operatorname{Re}(e^{\lambda_1 t} \mathbf{v}_1) + c_2 \operatorname{Im}(e^{\lambda_1 t} \mathbf{v}_1).$$

The equilibrium point is $\mathbf{a} = \mathbf{0}$ and may have various other properties aside from its stability as well.

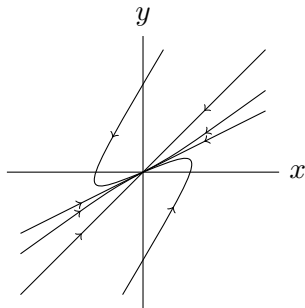
Stable node

In the case where $\lambda_1 < \lambda_2 < 0$,

1. Draw lines representing the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of λ_1 and λ_2 .
2. Construct trajectories starting far from the equilibrium point.
3. The trajectory starts close to the direction of $\mathbf{v}_1$, but approaches $\mathbf{v}_2$ as it approaches the equilibrium point.

When $\lambda_1 = -3$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\lambda_2 = -1$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$,
solutions have the form

$$\mathbf{x} = Ae^{-3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + Be^{-t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$



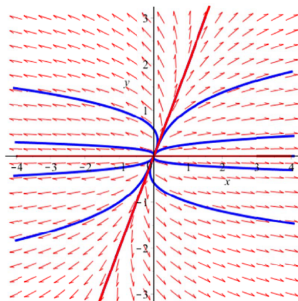
Unstable node

In the case where $0 < \lambda_1 < \lambda_2$,

1. Draw lines representing the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of λ_1 and λ_2 .
2. Construct trajectories starting from the equilibrium point.
3. Initially, the trajectory is close to $\mathbf{v}_1$, but approaches the direction of $\mathbf{v}_2$ quickly.
4. All trajectories are directed away from the equilibrium point.

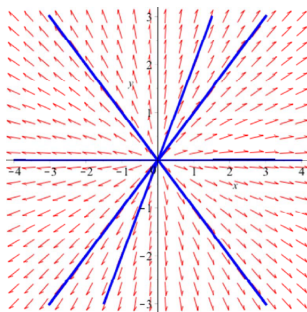
When $\lambda_1 = 1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\lambda_2 = 2, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, solutions have the form

$$\mathbf{x} = Ae^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + Be^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$



Unstable star

Here $\lambda_1 = \lambda_2 \neq 0$ and all nonzero vectors are eigenvectors. Therefore, all trajectories are either being attracted or repelled by the equilibrium points in a straight line.



There is inward attraction if $\lambda_1 < 0$, outward repulsion if $\lambda_1 > 0$.

Unstable saddle node

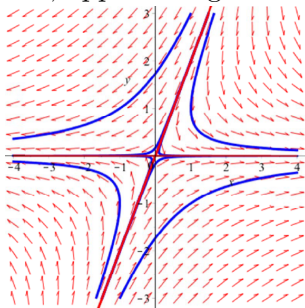
Here, $\lambda_1 < 0 < \lambda_2$.

1. Draw lines representing the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of λ_1 and λ_2 respectively. Trajectories on $\mathbf{v}_1$ and $\mathbf{v}_2$ are straight lines. Those on $\mathbf{v}_1$ are attracted towards the equilibrium point, whereas those on $\mathbf{v}_2$ are repelled.
2. Draw all other trajectories starting close to $\mathbf{v}_1$, initially approaching the equilibrium point. Such trajectories are eventually repelled from the equilibrium point, approaching $\mathbf{v}_2$.

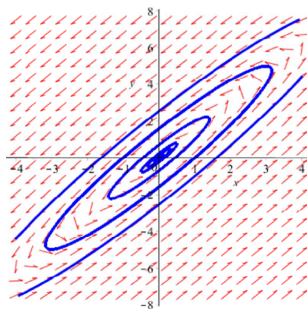
When $\lambda_1 = -1$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\lambda_2 = 1$,

$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, solutions have the form

$$\mathbf{x} = Ae^{-t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + Be^t \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

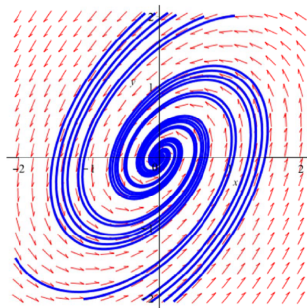


Here λ_1, λ_2 are purely imaginary. If eigenvalues are purely imaginary, the trajectory forms orbits around the equilibrium point as, in the general solution, we can rewrite e^{it} terms as $\cos(t) + i \sin(t)$. We can determine whether the orbit is clockwise or anticlockwise by finding the value of $d\mathbf{x}/dt$ at a point.



Stable foci

Here $\lambda_1 = \alpha + i\beta = \bar{\lambda}_2$. The trajectory will spiral inward when $\alpha < 0$ or outward when $\alpha > 0$. We can determine whether the spiral is clockwise or anticlockwise by finding the value of $d\mathbf{x}/dt$ at a point.



Equilibrium points example

Example 13: MATH2221 2019 2(i), parts (a) and (b)

Consider the following linear system of equations

$$\begin{aligned}\frac{dx}{dt} &= x + 2y, \\ \frac{dy}{dt} &= -5x - y.\end{aligned}$$

- (a) Find the general solution to the above system of equations.
- (b) Determine the nature and stability of the equilibrium point $(0, 0)$.

Equilibrium points example

Example 13: MATH2221 2019 2(i), parts (a) and (b)

Consider the following linear system of equations

$$\begin{aligned}\frac{dx}{dt} &= x + 2y, \\ \frac{dy}{dt} &= -5x - y.\end{aligned}$$

- (a) Find the general solution to the above system of equations.
- (b) Determine the nature and stability of the equilibrium point $(0, 0)$.

As a start, this system can be written as

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x} = \begin{pmatrix} 1 & 2 \\ -5 & -1 \end{pmatrix} \mathbf{x}.$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i, \lambda_2 = -3i$.

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i$, $\lambda_2 = -3i$. Finding an eigenvector for λ_1 is annoying, but there is a nice trick:

$$(A - 3iI)\mathbf{v} = \mathbf{0}$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i$, $\lambda_2 = -3i$. Finding an eigenvector for λ_1 is annoying, but there is a nice trick:

$$(A - 3iI)\mathbf{v} = \mathbf{0} \implies \begin{pmatrix} 1 - 3i & 2 \\ -5 & -1 - 3i \end{pmatrix} \mathbf{v} = \mathbf{0}$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i$, $\lambda_2 = -3i$. Finding an eigenvector for λ_1 is annoying, but there is a nice trick:

$$\begin{aligned}(A - 3iI)\mathbf{v} = \mathbf{0} &\implies \begin{pmatrix} 1 - 3i & 2 \\ -5 & -1 - 3i \end{pmatrix} \mathbf{v} = \mathbf{0} \\ &\implies (1 - 3i)v_1 + 2v_2 = 0\end{aligned}$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i$, $\lambda_2 = -3i$. Finding an eigenvector for λ_1 is annoying, but there is a nice trick:

$$\begin{aligned}(A - 3iI)\mathbf{v} = \mathbf{0} &\implies \begin{pmatrix} 1 - 3i & 2 \\ -5 & -1 - 3i \end{pmatrix} \mathbf{v} = \mathbf{0} \\ &\implies (1 - 3i)v_1 + 2v_2 = 0 \\ &\implies v_2 = \frac{-1 + 3i}{2}v_1\end{aligned}$$

Equilibrium points example

The eigenvalues λ_1, λ_2 satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A) = 0 \quad \text{and} \quad \lambda_1 \lambda_2 = \det(A) = 9.$$

Solving, we obtain eigenvalues $\lambda_1 = 3i$, $\lambda_2 = -3i$. Finding an eigenvector for λ_1 is annoying, but there is a nice trick:

$$\begin{aligned}(A - 3iI)\mathbf{v} = \mathbf{0} &\implies \begin{pmatrix} 1 - 3i & 2 \\ -5 & -1 - 3i \end{pmatrix} \mathbf{v} = \mathbf{0} \\ &\implies (1 - 3i)v_1 + 2v_2 = 0 \\ &\implies v_2 = \frac{-1 + 3i}{2}v_1 \\ &\implies \mathbf{v} = \begin{pmatrix} 2 \\ -1 + 3i \end{pmatrix} v_1.\end{aligned}$$

Equilibrium points example

The general solution to this system is therefore

$$\mathbf{x}(t) = c_1 \operatorname{Re} \left\{ e^{3it} \begin{pmatrix} 2 \\ -1 + 3i \end{pmatrix} \right\} + c_2 \operatorname{Im} \left\{ e^{3it} \begin{pmatrix} 2 \\ -1 + 3i \end{pmatrix} \right\}.$$

Equilibrium points example

The general solution to this system is therefore

$$\mathbf{x}(t) = c_1 \operatorname{Re} \left\{ e^{3it} \begin{pmatrix} 2 \\ -1 + 3i \end{pmatrix} \right\} + c_2 \operatorname{Im} \left\{ e^{3it} \begin{pmatrix} 2 \\ -1 + 3i \end{pmatrix} \right\}.$$

By writing $e^{3it} = \cos 3t + i \sin 3t$, one can identify these real and imaginary parts to find that the general solution is in fact

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 2 \cos 3t \\ -\cos 3t - 3 \sin 3t \end{pmatrix} + c_2 \begin{pmatrix} 2 \sin 3t \\ -\sin 3t + 3 \cos 3t \end{pmatrix}.$$

Since the eigenvalues were in this case purely imaginary, the equilibrium point $(0, 0)$ is a stable centre.

First integrals (2221 only)

Nonlinear problems are much harder to solve. Nevertheless, there is a simple (in principle) method for solving some of them based on the following concept.

Definition 17

A function $G : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **first integral** for a system of ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x})$$

if $G(\mathbf{x}(t))$ is constant for every solution $\mathbf{x}(t)$.

Geometrically, G is a first integral iff

$$\nabla G(\mathbf{x}) \perp \mathbf{F}(\mathbf{x}) \text{ for all } \mathbf{x}.$$

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

(Hint: find dy/dx first.)

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

(Hint: find dy/dx first.)

We can write

$$\frac{dy}{dx}$$

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

(Hint: find dy/dx first.)

We can write

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

(Hint: find dy/dx first.)

We can write

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{(x-1)y}{x(1-y)}$$

First integrals (2221 only)

Example 14

Find a first integral G of the system of nonlinear ODEs

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} x - xy \\ xy - y \end{pmatrix}.$$

Check your answer by computing $\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x})$.

(Hint: find dy/dx first.)

We can write

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{(x-1)y}{x(1-y)} \implies \left(\frac{1}{y} - 1 \right) dy = \left(1 - \frac{1}{x} \right) dx.$$

First integrals (2221 only)

Integrating both sides gives

$$\ln|y| - y = x - \ln|x| + c$$

First integrals (2221 only)

Integrating both sides gives

$$\ln|y| - y = x - \ln|x| + c \implies \ln|xy| - (x + y) = c$$

First integrals (2221 only)

Integrating both sides gives

$$\begin{aligned}\ln|y| - y = x - \ln|x| + c &\implies \ln|xy| - (x + y) = c \\ &\implies xye^{-x-y} = C.\end{aligned}$$

First integrals (2221 only)

Integrating both sides gives

$$\begin{aligned}\ln|y| - y = x - \ln|x| + c &\implies \ln|xy| - (x + y) = c \\ &\implies xye^{-x-y} = C.\end{aligned}$$

So, a first integral for the system is $G(x, y) = xye^{-x-y}$.

First integrals (2221 only)

Integrating both sides gives

$$\begin{aligned}\ln|y| - y = x - \ln|x| + c &\implies \ln|xy| - (x + y) = c \\ &\implies xye^{-x-y} = C.\end{aligned}$$

So, a first integral for the system is $G(x, y) = xye^{-x-y}$. As a sanity check, we have

$$\begin{aligned}\nabla G(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x}) &= -e^{-x-y} \begin{pmatrix} (x-1)y \\ x(y-1) \end{pmatrix} \cdot \begin{pmatrix} x-xy \\ xy-y \end{pmatrix} \\ &= -e^{-x-y} [-xy(x-1)(y-1) + xy(x-1)(y-1)] \\ &= 0.\end{aligned}$$

Thank you all for coming!

Please scan the QR code below and fill out the linked feedback form to give some feedback on today's revision seminar!

